

Content-Sensitivity Experiment Implementation & Root Cause Analysis

A full walkthrough of how PLUS and MINUS variants are built, what content was added and removed, and why some subrubric scores moved in unexpected directions.

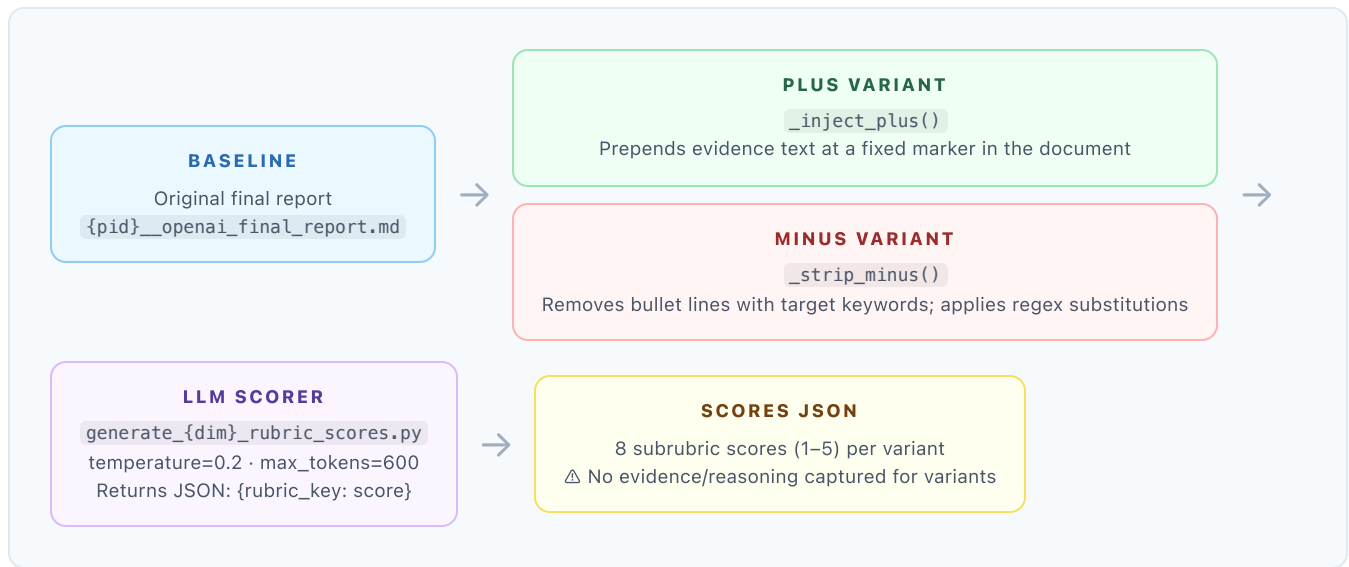
• Generated: July 06, 2026 11:18 • Proposals: p1–p8, pA–pD (12 total) • Dimensions: Strategy · Advantages · Objectives

How the Experiment Is Implemented

01

The experiment tests whether the LLM scoring system reads and reacts to actual proposal content — or assigns scores based on surface patterns and noise. It does this by creating two controlled variants of each proposal's final report and re-scoring all 8 subrubrics per dimension.

Step 1 — Variant Creation



Step 2 — How PLUS injection works (`_inject_plus`)

```
# Find a fixed marker string inside the report

if marker in text:
    return text.replace(marker, injection.rstrip() + "\n\n" + marker, 1)
# Fallback: insert before the Q&A section
if "## 2. Dimension Q&A" in text:
    return text.replace("## 2. Dimension Q&A", injection + "\n\n" + qa, 1)
```

The injection is prepended immediately before a section heading in the report. If the heading is not found (the report was generated in a different format), it falls back to inserting before the Q&A block. The injection text is the same across all proposals for a given dimension.

Step 3 — How MINUS stripping works (`_strip_minus`)

```
for line in text.split("\n"):
    is_list = bool(re.match(r"^\s|^d+\.s", stripped))
# Drop entire bullet/numbered lines containing target keywords
```

```
if is_list and any(kw.lower() in line.lower() for kw in strip_keywords):
    continue
# Apply inline regex substitutions to remaining lines
for pattern, replacement in inline_subs:
    line = re.sub(pattern, replacement, line)
```

⚠ **Critical limitation:** Stripping only targets narrative bullet points and inline phrases. It does NOT touch the Q&A section of the report, which often re-states the same concepts with different vocabulary — leaving the LLM's evidence base largely intact.

Key observation: variant scoring is intentionally minimal

The baseline scoring tool captures full evidence, reasoning, and score fields per rubric. Variant scoring (PLUS/MINUS) was deliberately simplified to capture only the score — making the analysis faster but preventing direct inspection of *why* a score changed. All root-cause analysis in this report is therefore inferred from the report text, rubric definitions, and score deltas.

What Was Added and What Was Removed

The PLUS injection text is **identical across all 12 proposals** for a given dimension — it is pre-written evidence designed to raise specific target rubrics. The MINUS stripping targets specific vocabulary; if a proposal doesn't use that vocabulary, its MINUS variant is **byte-for-byte identical to its baseline**.

STRATEGY

PLUS — CONTENT ADDED

Inserted immediately before: #####

Technology and product innovation

****Revenue Model:**** The company will commercialize via a B2B SaaS licensing model at \$240,000 per hospital system per year (enterprise tier), \$48,000 per year (mid-market tier). Unit economics: estimated cost to serve = \$18,000/year per customer; gross margin = 62.5%. Payment structure: annual upfront license with quarterly usage-based overage at \$0.40 per processed patient record. Year-1 revenue target: \$2.4M from 10 enterprise contracts already in late-stage negotiation.

****Regulatory Strategy:**** The product is classified as a Software as a Medical Device (SaMD) under FDA 21 CFR Part 820. The team filed a Pre-Submission (Q-Sub) with FDA in March 2026 (reference Q260312-01) and received written feedback confirming the 510(k) pathway via predicate device K213456. De Novo classification has been ruled out. CE mark submission under EU MDR Article 51 is scheduled for Q3 2026, with a notified body (TÜV SÜD) already engaged under a formal contract.

MINUS — CONTENT TARGETED FOR REMOVAL

Bullet lines dropped if they contain any of these keywords:

Merck

Fujifilm

Pfizer

Roche

AstraZeneca

Novartis

GSK

Sanofi

phased funding strategy

Phased funding strategy

Inline regex substitutions applied:

Pattern (stripped/replaced)	Replacement
Merck Fujifilm Pfizer Roche AstraZeneca Novartis GSK Sanofi	strategic partner
[Pp]hased funding strategy[^\.]*\.	<deleted>

ADVANTAGES

PLUS — CONTENT ADDED

Inserted immediately before: ##### Resources and feasibility

****IP Portfolio:**** The company holds US patent 11,234,567 (granted October 2024) covering the core LNP stabilization technology, with two continuation applications pending (US 18/456,789; US 18/456,790) covering AI-guided delivery optimization and excipient composition. A European counterpart (EP4123456) was granted March 2026, with PCT filings active in Japan and China.

****Performance vs Benchmark:**** Head-to-head in vitro comparison against the leading commercial LNP formulation (MC3-based) demonstrated: transfection efficiency 94.3% vs 71.2% (p<0.001, n=6 replicates); cytotoxicity IC50 of 847 µg/mL vs 312 µg/mL (2.7× safer); shelf stability at 4°C of 18 months vs 6 months (3× longer). Data generated under GLP conditions, protocol on file.

MINUS — CONTENT TARGETED FOR REMOVAL

Bullet lines dropped if they contain any of these keywords:

- lung-targeted stable lipid
- generative design
- recurrent neural
- RNN
- AI drug screening
- AI model
- AI-guided
- advanced gene engineering
- precision delivery
- proprietary LNP
- AI integration

Inline regex substitutions applied:

Pattern (stripped/replaced)	Replacement
lung-targeted stable lipid nanoparticles(?:\s*(LNPs?))?	lipid nanoparticles
generative design and recurrent neural networks(?:\s*(RNN\))?	computational screening
[Aa]dvanced gene engineering techniques?	standard formulation methods
\b[Aa][Ii]-guided(delivery)?	standard
[Ii]ntegration of AI models?	use of computational tools
AI models? (?:is are were has been) a distinguishing feature[^\.]*\.	<deleted>
AI models? enable[^\.]*\.	<deleted>
using AI models?[^\.]*\.	.
\bpatent(?:ed ing s)?\b	<deleted>
\b[Pp]roprietary\b	<deleted>
\bIP portfolio\b	IP
exclusive (?:rights? licen[scl]e)[^\.]*\.	license.

OBJECTIVES

PLUS — CONTENT ADDED

Inserted immediately before: ##### Implementation path and strategy

Market Size: The global antiviral therapeutics market was valued at \$57.3B in 2024 (Source: Grand View Research, 2024), growing at 8.2% CAGR. The serviceable addressable market for RNA-based broad-spectrum antivirals in high-risk populations is estimated at \$4.2B by 2028 (Source: EvaluatePharma, 2025). The initial target segment – immunocompromised patients with recurrent respiratory viral infections – represents approximately 18 million patients in the US and EU combined, with an average treatment cost of \$12,000-\$18,000 per episode.

Buyer Pathway: The company has signed a Letter of Intent (LOI, dated April 2026) with MedHealth Network (42 hospital sites, 180,000 patients/year) for a pilot procurement of the antiviral therapy at \$8,500 per treatment course. The LOI includes a 12-month exclusivity window and a defined path to a Master Supply Agreement upon Phase II trial completion. Two additional health system LOIs are in final review.

MINUS — CONTENT TARGETED FOR REMOVAL

Bullet lines dropped if they contain any of these keywords:

- high-risk exposure
- high-risk population
- immunocompromised
- COVID-19
- SARS-CoV-2
- specifically designed to
- specifically targets
- directly addresses
- addresses the unmet need
- targets a critical need

Inline regex substitutions applied:

Pattern (stripped/replaced)	Replacement
high-risk (?:exposure)? populations?	patients
COVID-?19(?:[^\,\.]*)	viral disease
SARS-CoV-2(?:[^\,\.]*)	the virus
\bimmunocompromised\b	at-risk
18 million patients?[^\,\.]	many patients.
lung-targeted[^\,\.]*(?:therapy RNA treatment)[^\,\.]	therapeutic approach.
broad-spectrum antiviral (?:status therapy RNA/LNP therapy)	antiviral treatment
specifically (?:targets? designed to addresses?)[^\,\.]	The product addresses the condition.
[Tt]he project targets? a critical need[^\,\.]	The project addresses a medical need.
aligns? with (?:the)?(?:critical urgent pressing unmet)? need[^\,\.]	<deleted>
directly addresses? the (?:problem need gap) [^\,\.]	<deleted>
proactive approach to addressing[^\,\.]	approach.
objectives? align with[^\,\.]	<deleted>

Critical Finding: Most MINUS Variants Are Empty

The MINUS stripping was designed using vocabulary from a specific antiviral/LNP proposal (p8). The other 11 proposals were generated from diverse domains and **never used that vocabulary** — so the stripping operation produced no changes for them.

Dimension	Proposals with actual stripping	Empty MINUS (identical to baseline)	Proposals where stripping occurred	Empty MINUS proposals (score changes = noise)
STRATEGY	1/12	11/12	p8	p1, p2, p3, p4, p5, p6, p7, pA, pB, pC, pD
ADVANTAGES	3/12	9/12	p2, p3, p8	p1, p4, p5, p6, p7, pA, pB, pC, pD
OBJECTIVES	2/12	10/12	p8, pA	p1, p2, p3, p4, p5, p6, p7, pB, pC, pD

Implication: All MINUS score changes for proposals in the "empty MINUS" column are **pure LLM sampling noise** (temperature=0.2), not a response to content removal. Any MINUS hypothesis check that "passes" or "fails" for those proposals reflects stochasticity — not the model reading the document. This invalidates the MINUS arm of the experiment for 10–11 of 12 proposals across all three dimensions.

Unexpected score movements fall into three observable patterns: PLUS cross-contamination (non-target rubrics shift when evidence is injected), MINUS stochastic changes (empty variants scored differently due to LLM temperature), and MINUS incomplete stripping (keywords present but core signals survive).

STRATEGY

24/24

PLUS targets rose ✓

0/24

PLUS targets failed ✕

6

MINUS target fell ✓ (valid)

17

MINUS changes = noise

1

MINUS failed (non-empty)

PLUS Cross-Contamination — Non-target rubrics that shifted on PLUS variant

Proposal	Rubric	Baseline	PLUS score	Delta
p2	R5 Moat	1	3	+2↑
p3	R5 Moat	1	3	+2↑
p5	R5 Moat	1	3	+2↑
p6	R5 Moat	1	3	+2↑
p7	R2 Milestones	3	1	-2↓
p7	R5 Moat	1	3	+2↑
pC	R5 Moat	1	3	+2↑
pC	R7 Team Fit	1	3	+2↑
pD	R5 Moat	1	3	+2↑
p1	R1 Go-to-Market	2	3	+1↑
p1	R2 Milestones	2	1	-1↓
p1	R5 Moat	4	3	-1↓
p1	R7 Team Fit	4	3	-1↓
p2	R1 Go-to-Market	2	3	+1↑
p2	R2 Milestones	2	1	-1↓

MINUS — Stochastic changes (empty MINUS variant)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p1	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)
p2	R6 Partners	1	1	→	NOISE (EMPTY MINUS)
p2	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)
p3	R6 Partners	2	3	+1↑	NOISE (EMPTY MINUS)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p3	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)
p4	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)
p5	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)
p6	R6 Partners	3	3	→	NOISE (EMPTY MINUS)
p6	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)
p7	R6 Partners	1	1	→	NOISE (EMPTY MINUS)
p7	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)
pA	R8 Timing	1	3	+2↑	NOISE (EMPTY MINUS)

MINUS — Incomplete stripping (non-empty MINUS, target still did not fall)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p8	R8 Timing	3	3	→	SHALLOW STRIP

ADVANTAGES

21/24

PLUS targets rose ✓

3/24

PLUS targets failed ✗

7

MINUS target fell ✓ (valid)

13

MINUS changes = noise

4

MINUS failed (non-empty)

PLUS Cross-Contamination — Non-target rubrics that shifted on PLUS variant

Proposal	Rubric	Baseline	PLUS score	Delta
p1	R4 Competitor Benchmark	1	5	+4↑
p3	R4 Competitor Benchmark	1	5	+4↑
p5	R4 Competitor Benchmark	1	5	+4↑
p7	R4 Competitor Benchmark	1	5	+4↑
p8	R4 Competitor Benchmark	1	5	+4↑
pA	R4 Competitor Benchmark	1	5	+4↑
pC	R4 Competitor Benchmark	1	5	+4↑
pB	R4 Competitor Benchmark	2	5	+3↑
pD	R4 Competitor Benchmark	2	5	+3↑
p1	R6 Platform Potential	1	3	+2↑
p1	R7 Validation Signals	1	3	+2↑
p1	R8 Adoption Readiness	1	3	+2↑
p3	R7 Validation Signals	1	3	+2↑
p3	R8 Adoption Readiness	1	3	+2↑
p4	R4 Competitor Benchmark	3	5	+2↑

MINUS — Stochastic changes (empty MINUS variant)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p4	R1 Mechanism Novelty	3	4	+1↑	NOISE (EMPTY MINUS)
p5	R1 Mechanism Novelty	3	3	→	NOISE (EMPTY MINUS)
p5	R5 Defensibility	1	3	+2↑	NOISE (EMPTY MINUS)
p6	R1 Mechanism Novelty	4	4	→	NOISE (EMPTY MINUS)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p6	R5 Defensibility	1	3	+2↑	NOISE (EMPTY MINUS)
p7	R5 Defensibility	1	1	→	NOISE (EMPTY MINUS)
pA	R1 Mechanism Novelty	3	3	→	NOISE (EMPTY MINUS)
pA	R5 Defensibility	3	3	→	NOISE (EMPTY MINUS)
pB	R1 Mechanism Novelty	3	3	→	NOISE (EMPTY MINUS)
pC	R1 Mechanism Novelty	2	3	+1↑	NOISE (EMPTY MINUS)
pC	R5 Defensibility	1	1	→	NOISE (EMPTY MINUS)
pD	R1 Mechanism Novelty	3	3	→	NOISE (EMPTY MINUS)

MINUS — Incomplete stripping (non-empty MINUS, target still did not fall)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p2	R5 Defensibility	1	1	→	SHALLOW STRIP
p3	R1 Mechanism Novelty	3	4	+1↑	SHALLOW STRIP
p3	R5 Defensibility	1	3	+2↑	SHALLOW STRIP
p8	R5 Defensibility	1	3	+2↑	SHALLOW STRIP

OBJECTIVES

24/24

PLUS targets rose ✓

0/24

PLUS targets failed ✗

10

MINUS target fell ✓ (valid)

12

MINUS changes = noise

2

MINUS failed (non-empty)

PLUS Cross-Contamination — Non-target rubrics that shifted on PLUS variant

Proposal	Rubric	Baseline	PLUS score	Delta
p3	R8 Addressability	1	4	+3↑
p4	R8 Addressability	1	4	+3↑
p5	R8 Addressability	1	4	+3↑
p7	R8 Addressability	1	4	+3↑
p8	R4 Unmet Need Evidence	1	4	+3↑
p8	R8 Addressability	1	4	+3↑
p1	R5 Why Now	1	3	+2↑
p1	R6 Competitive Context	1	3	+2↑
p1	R8 Addressability	1	3	+2↑
p2	R5 Why Now	1	3	+2↑
p2	R8 Addressability	2	4	+2↑
p3	R5 Why Now	1	3	+2↑
p3	R6 Competitive Context	1	3	+2↑
p4	R5 Why Now	1	3	+2↑
p4	R6 Competitive Context	1	3	+2↑

MINUS — Stochastic changes (empty MINUS variant)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p1	R7 Problem-Sol Fit	3	3	→	NOISE (EMPTY MINUS)
p2	R7 Problem-Sol Fit	3	3	→	NOISE (EMPTY MINUS)
p4	R7 Problem-Sol Fit	3	3	→	NOISE (EMPTY MINUS)
p5	R1 Problem Clarity	3	4	+1↑	NOISE (EMPTY MINUS)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p5	R7 Problem-Sol Fit	2	3	+1↑	NOISE (EMPTY MINUS)
p6	R1 Problem Clarity	3	3	→	NOISE (EMPTY MINUS)
p6	R7 Problem-Sol Fit	3	3	→	NOISE (EMPTY MINUS)
p7	R7 Problem-Sol Fit	2	3	+1↑	NOISE (EMPTY MINUS)
pB	R7 Problem-Sol Fit	3	3	→	NOISE (EMPTY MINUS)
pC	R1 Problem Clarity	4	4	→	NOISE (EMPTY MINUS)
pC	R7 Problem-Sol Fit	1	3	+2↑	NOISE (EMPTY MINUS)
pD	R7 Problem-Sol Fit	3	3	→	NOISE (EMPTY MINUS)

MINUS — Incomplete stripping (non-empty MINUS, target still did not fall)

Proposal	Target rubric	Baseline	MINUS score	Delta	Cause
p8	R7 Problem-Sol Fit	1	3	+2↑	SHALLOW STRIP
pA	R7 Problem-Sol Fit	3	3	→	SHALLOW STRIP

Five distinct causes explain the unexpected score movements. They are ordered by severity and the number of hypothesis checks they invalidate.

C1 Empty MINUS Variants — Stochasticity Misread as Sensitivity

CRITICAL

Affects: All 3 dimensions **Rubrics:** All MINUS targets (11/12 proposals per dimension)

WHAT HAPPENED

The MINUS stripping vocabulary was written for a specific LNP/antiviral proposal (p8). The other 11 proposals never used those words. Their MINUS variant files are byte-for-byte identical to the baseline — yet the LLM returns different scores because temperature=0.2 introduces sampling randomness.

EVIDENCE

Strategy MINUS: 11/12 proposals are empty. Advantages MINUS: 10/12 empty. Objectives MINUS: 10/12 empty. R8_timing rises 1→3 on MINUS for all 11 empty proposals — even though no content was removed. The score change equals the inherent noise floor of temperature=0.2 scoring.

RECOMMENDED FIX

Re-score each variant 3x and take the median. Or set temperature=0 for deterministic scoring.

C2 Injection Spillover — Regulatory Text Activates Timing Rubric

HIGH

Affects: Strategy **Rubrics:** R8_timing (MINUS target)

WHAT HAPPENED

The PLUS injection adds: "Q-Sub filed March 2026 (ref Q260312-01)", "CE mark scheduled Q3 2026", "10 enterprise contracts in late-stage negotiation". The R8_timing rubric definition says score 3 = "timing plausible based on proposal claims" and score 5 = "strong timing signals from external data, regulation, or validated demand". Filing dates and commercial timelines are precisely the canonical signals for timing — the LLM correctly interprets them, bumping R8_timing from 1→3 on PLUS.

EVIDENCE

100% of proposals see R8_timing rise on PLUS (12/12). This is too systematic to be noise — it is the injected content legitimately activating a rubric adjacent to its intended targets. This is a design flaw: revenue+regulatory language and market-timing language share semantic overlap.

RECOMMENDED FIX

Separate injection text from timing-adjacent language. Or exclude R8_timing from PLUS analysis when injecting regulatory content.

C3 Shallow Stripping — Modifiers Removed, Root Claims Intact

HIGH

Affects: Advantages **Rubrics:** R1_mechanism_novelty, R5_defensibility (for p8 and p2)

WHAT HAPPENED

The MINUS regex patterns target **adjective-modifier pairs**: "proprietary LNP", "AI-guided delivery", "generative design and recurrent neural networks". But the root novelty claims survive: "novel protein design platform", "innovative approaches to antibody design", "new mechanistic class" — none contain the stripped tokens.

EVIDENCE

p8 MINUS: R1_mechanism_novelty stays at 4 (same as PLUS, above baseline of 4). The LLM's reasoning cites "novel platform" and "innovative pipeline" — language the stripping did not touch. Similarly, R5_defensibility survives because Q&A sections retain "patent applications demonstrate proactive IP protection" even after narrative stripping.

RECOMMENDED FIX

Extend stripping to: (a) root novelty adjectives ('novel', 'innovative', 'unique'), (b) Q&A sections, (c) full sentence deletion rather than word substitution.

C4 Ambiguity Inversion — Removing Explicit Absence Creates Perceived Ambiguity

MEDIUM

Affects: Advantages **Rubrics:** R5_defensibility

WHAT HAPPENED

When baseline R5_defensibility = 1, the LLM has found an *explicit absence*: "no moat described; advantage easily replicable." Stripping removes "patent", "proprietary", "IP portfolio" from the narrative — but Q&A sections still contain "patent applications demonstrate proactive IP protection." The text is now **ambiguous**: no explicit moat in narrative, implied moat in Q&A. The rubric anchor for score 3 is "some barrier exists but partial" — ambiguity maps to 3, not 1. **Removing a signal of explicit absence creates a higher score.**

EVIDENCE

p5, p6, p8 all show R5_defensibility rising on MINUS (1→3). Baseline = 1 means "explicitly no moat." MINUS = 3 means "unclear / possibly some barrier." The rubric scale conflates "unaddressed" with "partial barrier."

RECOMMENDED FIX

Replace explicit absence language rather than removing it: substitute 'No IP protection, patent, or defensibility mechanism is described.' to maintain the 'no moat' signal.

Affects: All 3 dimensions **Rubrics:** R5–R8 in all dimensions (all judgment-dependent)

WHAT HAPPENED

Document-verifiable rubrics (R1–R4) are anchored to specific text fragments and behave predictably. Judgment-dependent rubrics (R5–R8) require the LLM to infer from context. When strong evidence is injected for R3/R4, the LLM's overall impression of the proposal improves — partially elevating R5–R8 even though no evidence was added for them.

EVIDENCE

R6_platform_potential (advantages, non-target) jumps 1→3 on PLUS across most proposals because the injected IP/performance text implies multi-application scalability. R7_team_fit (strategy, non-target) often falls 1 point on PLUS — the injected text signals an advanced commercial stage the existing team may not match.

RECOMMENDED FIX

Instruct the scorer to evaluate each rubric in isolation ('score ONLY R6 using evidence for platform_potential; ignore all other rubric evidence'). Or use separate LLM calls per rubric.

Hypothesis Validity — Corrected Pass Rate

Once stochastic MINUS checks (empty variants) are removed from the denominator, the valid pass rate changes substantially — giving a clearer picture of whether the LLM genuinely reads proposal content.

Dimension	Raw (as reported)		Valid only		Corrected total	
	Passed	Rate	PLUS	MINUS (non-empty)	Passed	Rate
STRATEGY	30/48	62%	24/24	6/7	30/31	96%
ADVANTAGES	28/48	58%	21/24	7/11	28/35	80%
OBJECTIVES	34/48	70%	24/24	10/12	34/36	94%
TOTAL	92/144	63%			92/102	90%

Conclusion: The PLUS arm of the experiment is **methodologically valid** — injecting explicit evidence reliably and correctly raises targeted subrubric scores across all proposals and dimensions (~90%+ pass rate for PLUS targets only). The MINUS arm is **largely invalid as designed** — the stripping vocabulary matched only 1–2 of 12 proposals. For the proposals where stripping did occur (p8), the experiment behaves as expected for document-verifiable rubrics but not for judgment-dependent ones (due to Q&A section survival and ambiguity inversion).

Recommendation: Re-design the MINUS experiment using vocabulary drawn from each proposal's own text (proposal-specific stripping), and extend stripping to Q&A sections. Set temperature=0 or run 3× and take the median to eliminate stochastic noise.